電腦視覺與深度學習
(Computer Vision and Deep Learning)

Homework 2

TA: 謝是奎

Gmail: nckubot65904@gmail.com

Office Hour: 10:00~12:00, Mon.

10:00~12:00, Thu.

At CSIE 9F Robotics Lab.

Notice (1/2)

- Copying homework is strictly prohibited!! Penalty: Both individuals will receive a score of 0!!
- Due date ➔ 09:00:00, 2026/01/02 (Fri.)

  Do not submit late, or the following points will be deducted:
  - Submit within seven days after the deadline, and your score will be reduced by half.
  - If you submit after this period, you will receive a score of 0.
- You must attend the demonstration, otherwise your score will be 0. The demonstration schedule will be announced on NCKU Moodle.
- You must create GUI, otherwise your point will be deducted.
- Upload to ➔ 140.116.154.28 → Upload/Homework/Hw2
  - User ID: cvdl2025 Password: RL2025cvdl
- Format
  - Filename: Hw2_StudentID_Name_Version.zip
    - Ex: Hw2_F71234567_林小明_V1.zip
    - If you want to update your file, you should update your version to be V2,
    - Ex: Hw2_F71234567_林小明_V2.zip
  - Content: Project folder *(Excluding the pictures)

    *Note: Remove your “Debug” folder to reduce file size.

Notice (2/2)

- Python (recommended):
  - Python 3.10
  - Matplotlib 3.8.0
  - UI framework: pyqt5 (5.15.11)
  - Torch 2.5.1
  - Torchvision 0.20.1
  - Torchsummary 1.5.1

Assignment Scoring (Total: 100%)

1. (50%) Train a Pascal VOC Object Detector using Faster R-CNN (出題：Allen)
   - 1.1 (15%) Load Model And Show Model Architecture.
   - 1.2 (15%) Show Training Loss Curve.
   - 1.3 (20%) Use The Trained Model to Run Inference.
2. (50%) Train a CIFAR-10 Classifier using ResNet18 (出題：Ian)
   - 2.1 (10%) Load and Show the Image
   - 2.2 (10%) Show the Architecture of ResNet18 Model
   - 2.3 (15%) Show Training/Validating Loss Curve and Accuracy Curve
   - 2.4 (15%) Use the Trained Model with Highest Validation Accuracy to Run Inference, Show the Probability Distribution and Class Label

1. Train a Pascal VOC Object Detector using Faster R-CNN (50%)

(出題：Allen)

1.1 (15%) Load Model And Show Model Architecture.

1.2 (15%) Show Training Loss Curve.

1.3 (20%) Use The Trained Model to Run Inference.

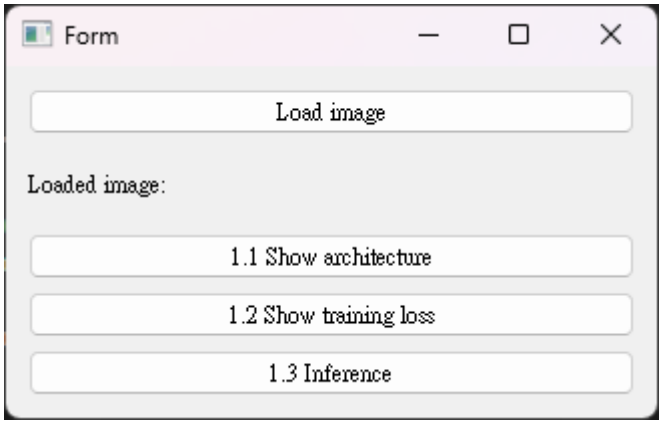
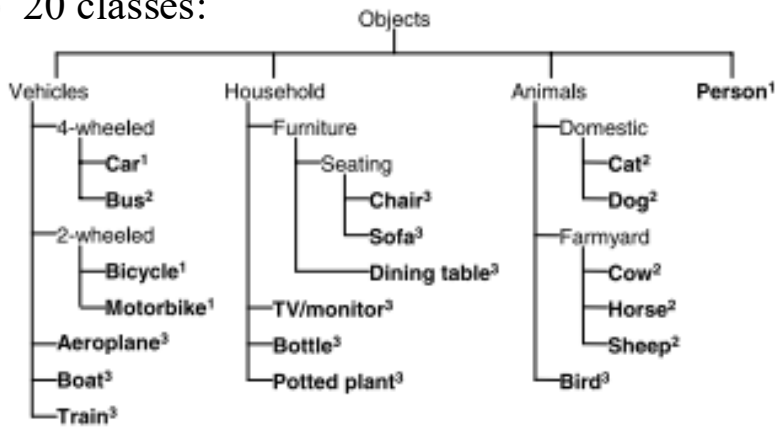


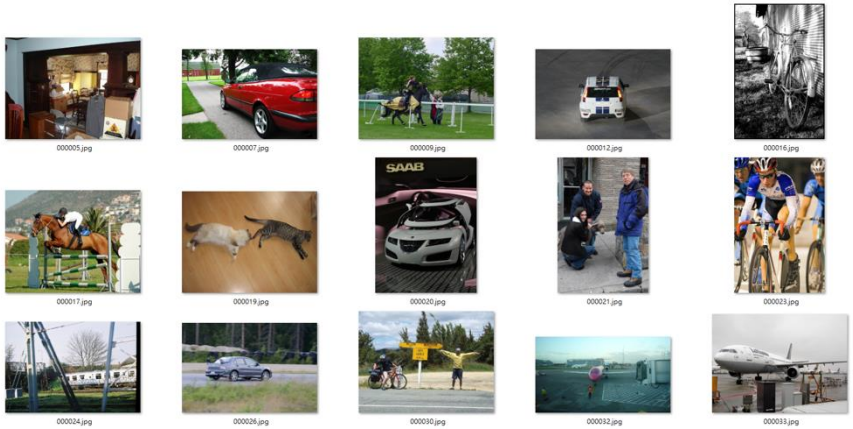
Fig.1 GUI example

1.0 Train a Pascal VOC Object Detector using Faster R-CNN (50%)

(出題：Allen)

1. Objective
   1) Learn how to train a Faster R-CNN model on the Pascal VOC dataset in order to detect objects within images and evaluate the model's detection accuracy.
2. Download Pascal VOC 2007 Dataset from FTP
   1) Data type: JPEG images + XML
   2) 20 classes:





3. Requirements
   1) Train Faster R-CNN model using PyTorch.
   2) In the submitted file, you need to include
      A. Figure of training loss in .jpg or .png format.
      B. Code for your GUI program
      C. Code for model training.
   3) Please do not include image data in the submitted file.

1.1 (15%) Load Model And Show Model Architecture. (1/2)

– FasterRCNN

(出題：Allen)

1) At home:

backbone 20 classes + __background__

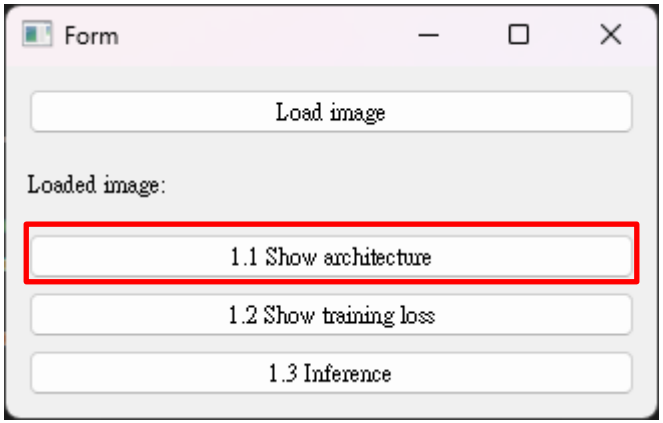
(1) Use model = torchvision.models.detection.fasterrcnn_resnet50_fpn (num_classes=21) to build the FasterRCNN model.

(2) Use print(model) to print the FasterRCNN architecture in the terminal.

2) When the demo:

(1) Click the button "1.1 Show architecture"

(2) Run the function to show the architecture in the terminal.



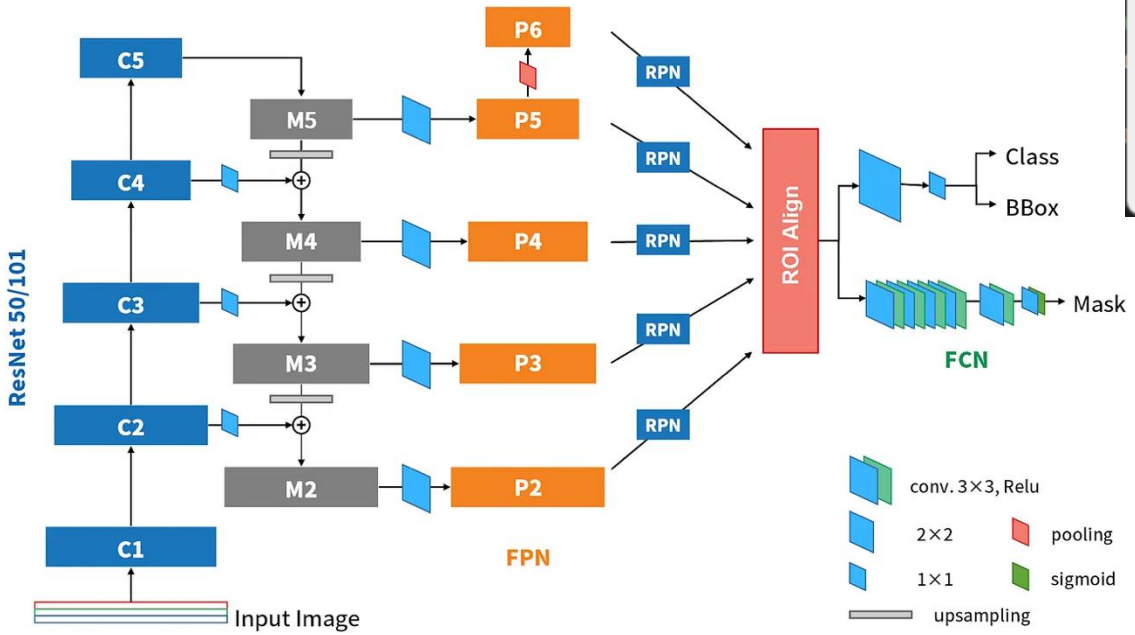


Fig.1 model architecture visualize

Reference: faserrcnn_resnet50_fpn

1.1 (15%) Load Model And Show Model Architecture. (2/2)

Code: (出題：Allen)

1. Convert VOC format (XML) → FasterRCNN expected format

```
root = "data\VOCtrainval_06-Nov-2007"
```

the folder path where PASCAL VOC dataset is stored

```
# load VOC trainval dataset
voc_trainval = VOCDetection(root, year="2007", image_set="trainval", download=False)
voc_trainval = VOCDatasetWrapper(voc_trainval, transforms=transform)

train_loader = DataLoader(voc_trainval, batch_size=4, shuffle=True, collate_fn=collate_fn)
```

A built-in PyTorch tool that automatically loads the raw PASCAL VOC images and their corresponding XML annotation files.

"VOCDatasetWrapper" converts XML into tensor that Faster R-CNN needs. (You need to implement it by yourself.)

"collate_fn" gathers data of different sizes (like images with different numbers of objects) into a single batch that the model can process.

2. Load model and show architecture

```
weights = FasterRCNN_ResNet50_FPN_Weights.COCO_V1
model = fasterrcnn_resnet50_fpn(weights=weights, num_classes=len(weights.meta["categories"]))
num_classes_voc = 21
in_features = model.roi_heads.box_predictor.cls_score.in_features
model.roi_heads.box_predictor.cls_score = torch.nn.Linear(in_features, num_classes_voc)
model.roi_heads.box_predictor.bbox_pred = torch.nn.Linear(in_features, num_classes_voc * 4)
```

pre-trained weights were trained on the COCO dataset.

get feature dimension

$(x_{min}, y_{min}, x_{max}, y_{max})$

```
print(model)
```

show model architecture

3. Parameters

```
initial_lr = 0.005
optimizer = torch.optim.SGD(
    model.parameters(),
    lr=initial_lr,
    momentum=0.9,
    weight_decay=0.0005
)
# 加入學習率排程器 (Scheduler)，每 8 個 Epoch 將 lr 降為 0.1 倍
scheduler = StepLR(optimizer, step_size=8, gamma=0.1)
```

1.2 (15%) Show Training Loss Curve

(出題：Allen)

1) At home:

(1) Load the training dataset.

(2) Train Q1.1 model at least 20 epochs at home and record the training loss in each epoch.

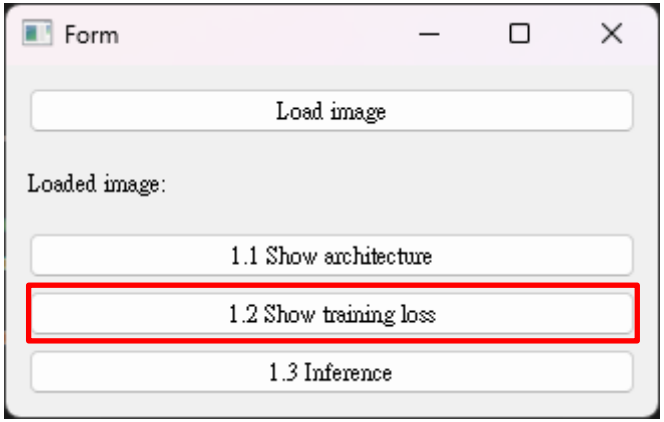
(3) Notice: If your loss is too high, you can try

A. Adjust the learning rate of the optimizer.

B. Change the data augmentation techniques used.

(4) Save weight file (.pth) with lowest loss. (how to save and load model in pytorch?)

(5) Use matplotlib.pyplot.plot() to create a line chart for the training loss and save the figure.



2) When the demo:

(1) Click the button "1.2 Show Training Loss".

(2) Show the saved figure of loss during training in a new window.

Notice: this is an example, the number of iterations might differ

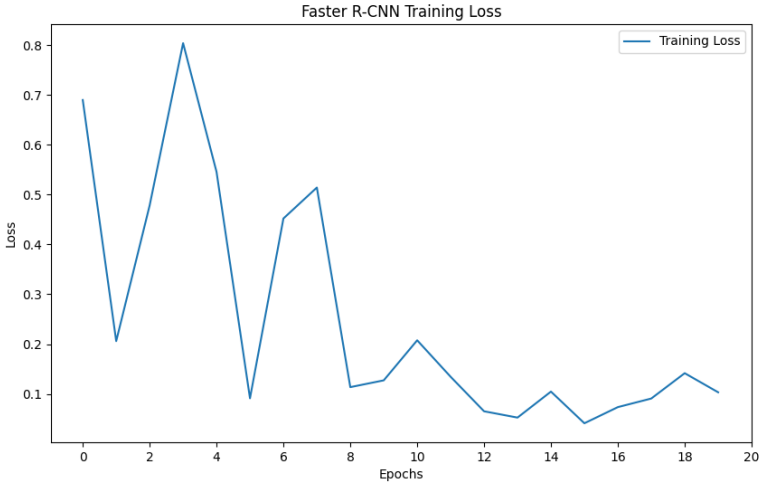


Fig.2 images for the loss during training

1.3 (20%) Use The Trained Model to Run Inference

(出題：Allen)

1) At home:

(1) Load the .pth model which trained at home. (how to save and load model in pytorch?)

(2) Click the button "1.3 Inference" to run inference on the image.

- Need to show bounding box(es), label(s) and confidence score(s) in the picture.

(3) Show the loaded image with bounding box and label on the GUI. (Fig.3)

2) When the demo: repeat the process

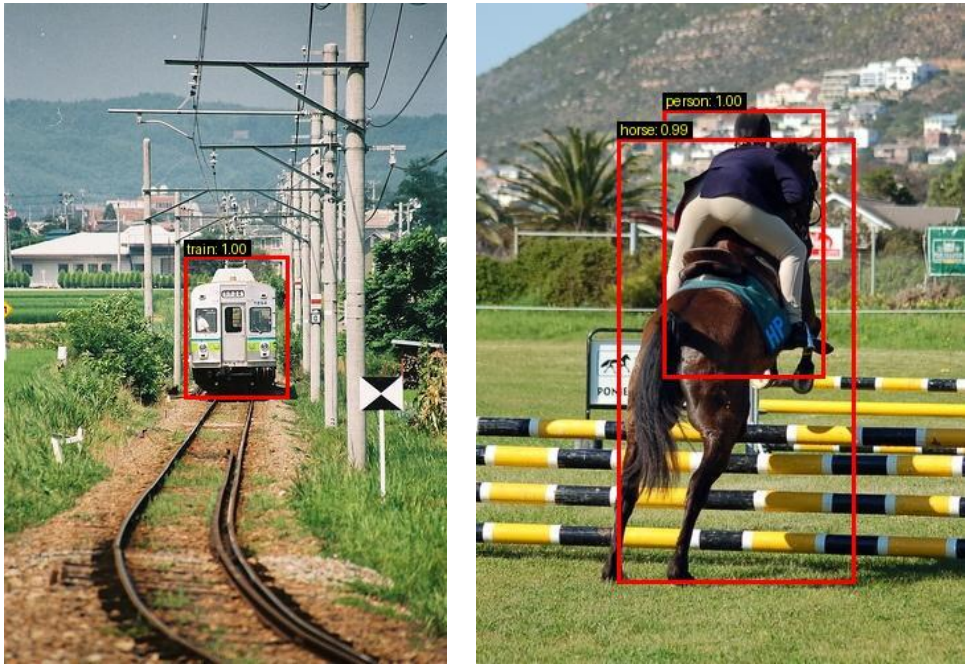
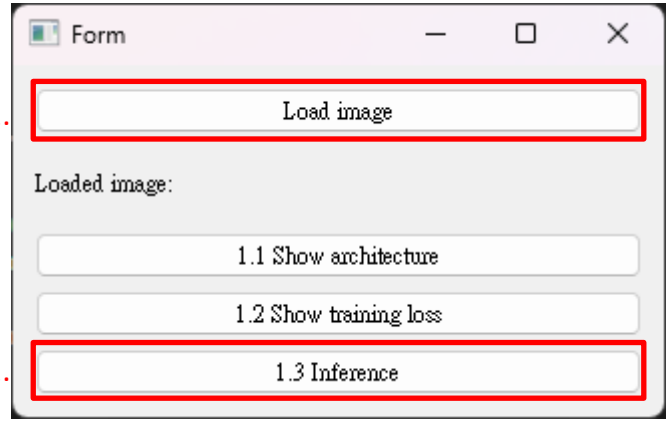
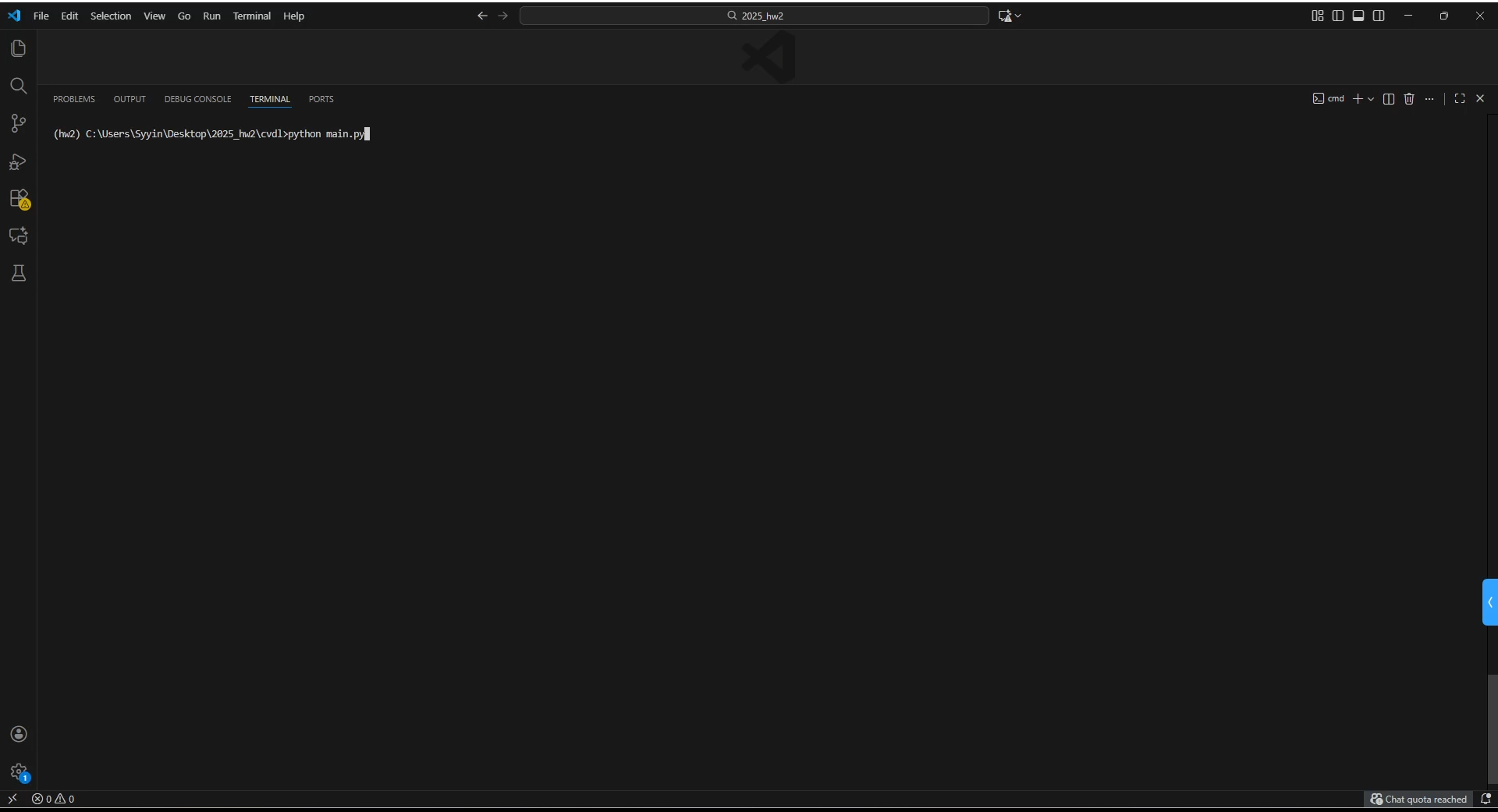


Fig.3 Inference



1. Training a Pascal VOC Object Detector using Faster R-CNN

(出題：Allen)



2. Train a CIFAR-10 Classifier Using ResNet18

(出題：Ian)

2.1 (10%) Load and Show the Image

2.2 (10%) Show the Architecture of ResNet18 Model

2.3 (15%) Show Training/Validating Loss Curve and Accuracy Curve

2.4 (15%) Use the Trained Model with Highest Validation Accuracy to Run Inference, Show the Probability Distribution and Class Label

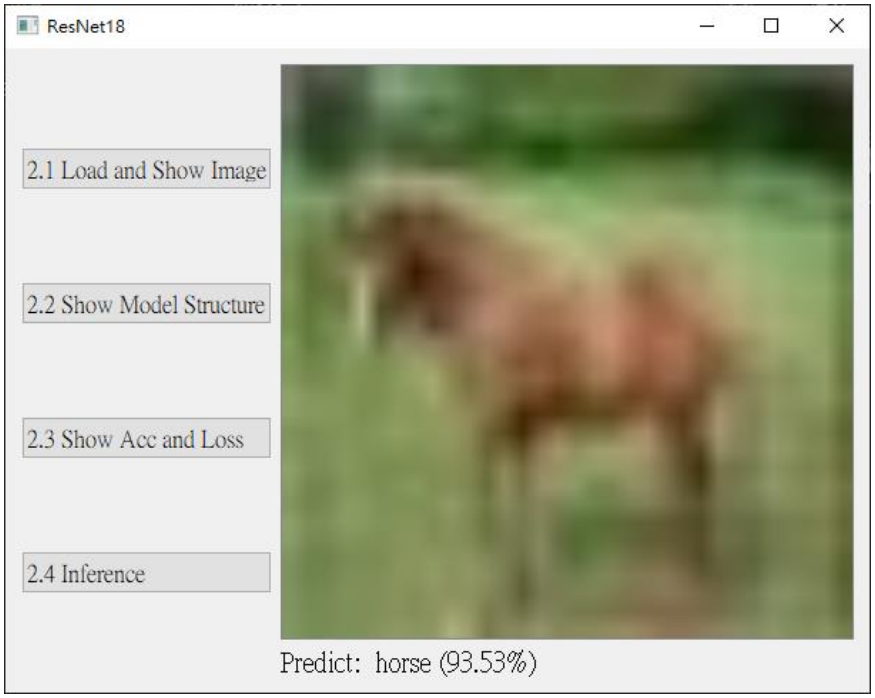


Figure: GUI Example

2.0 Train a CIFAR-10 Classifier Using ResNet18

(出題：Ian)

1. Objective
   1) Learn how to train a ResNet18 model to classify images of CIFAR-10 using PyTorch
2. Download CIFAR-10 Dataset
   1) Image size: 32x32x3c (RGB)
   2) 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck
   3) Training: 50,000 images in total
   4) Testing: 10,000 images in total
   5) Datasets (Download from torchvision.dataset)

I/P: Your data file path I/P: Set to true to load training data else load testing data

- torchvision.datasets.CIFAR10(data_path, download=True, train=True/False)

I/P: Download dataset or not

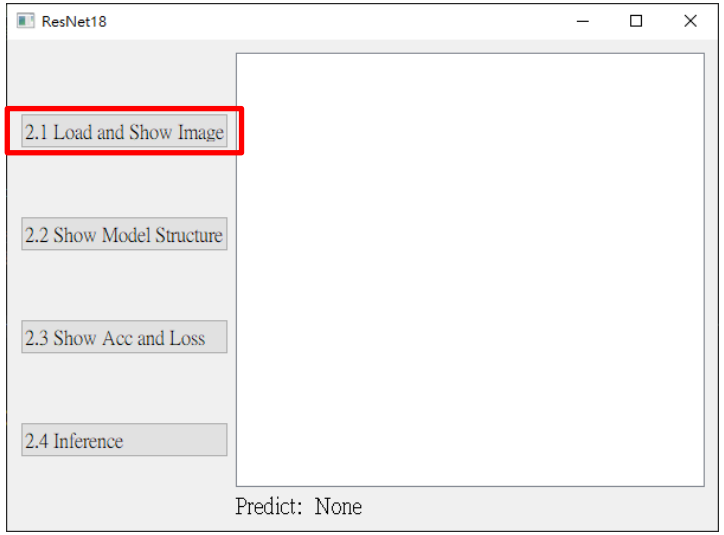
(1) Training and Validation dataset: You should split training and validation by yourself. (Suggestion: 80% and 20%)

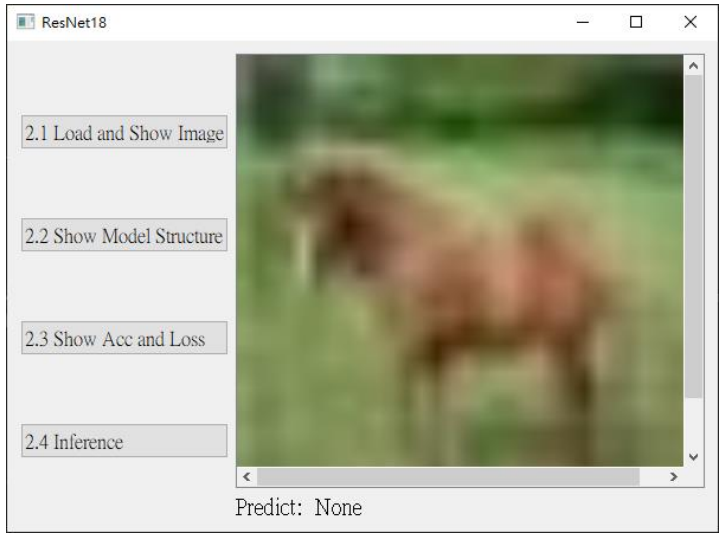
(2) Inference dataset: 10,000 images in total.
For demo: Download inference image (inference_img.zip) from FTP. There is a special image for demo.

2.1 (10%) Load and Show the Image

(出題：Ian)

1. At home:
   1) Click the button "2.1 Load and Show Image"
   2) Load an image
   3) Resize images to 32x32×3c (RGB)
   4) Show selected image on GUI
2. When the demo:
   1) Click the button "2.1 Load and Show Image"
   2) Load and show image on GUI





Notice: this is an example, the images might differ

2.2 (10%) Show the Architecture of ResNet18 Model

(出題：Ian)

1. At home:
   1) Replace 7x7 conv stride 2 with 3x3 conv stride 1 and remove max pooling to preserve feature information because CIFAR-10 image size is 32x32x3c (original input size is 224x224x3c)
   2) Replace the output layer to a FC (Fully Connected) layer of 10 node

      It's because that the node of FC Layer in ResNet18 is 1000.
      In our assignment, it only needs 10 nodes for 10 classes.
      As a result, we have to replace it.

      → Hint:

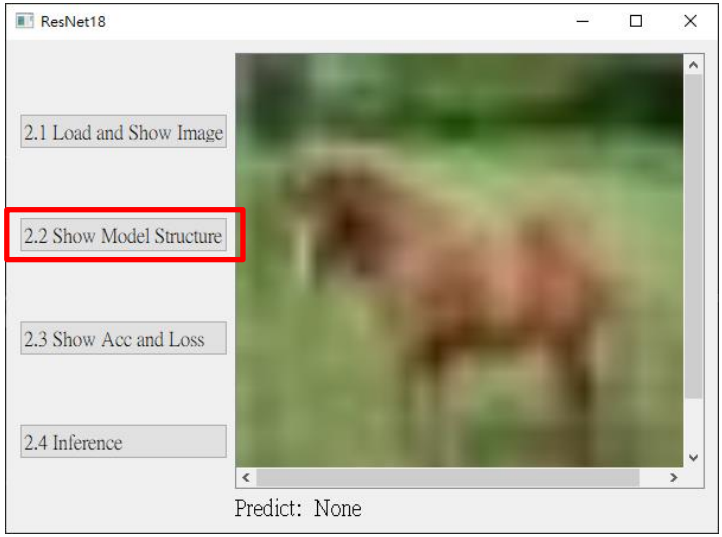
      I/P: Size of input feature map

      - torch.nn.Linear(512, 10) ➔ in order to change the original FC Layer of model.

      I/P: Size of output
   3) Run the function to show the structure in the terminal
2. When the demo:
   1) Click the button "2.2 Show Model Structure"
   2) Run the function to show the structure in the terminal

Layer Name	18-layer
conv1	7x7, 64, stride 2 3x3 max pool, stride 2
conv2_x	[3x3, 64; 3x3, 64] x 2
conv3_x	[3x3, 128; 3x3, 128] x 2
conv4_x	[3x3, 256; 3x3, 256] x 2
conv5_x	[3x3, 512; 3x3, 512] x 2
	average pool, 1000-d fc, softmax

Original ResNet-18 architecture

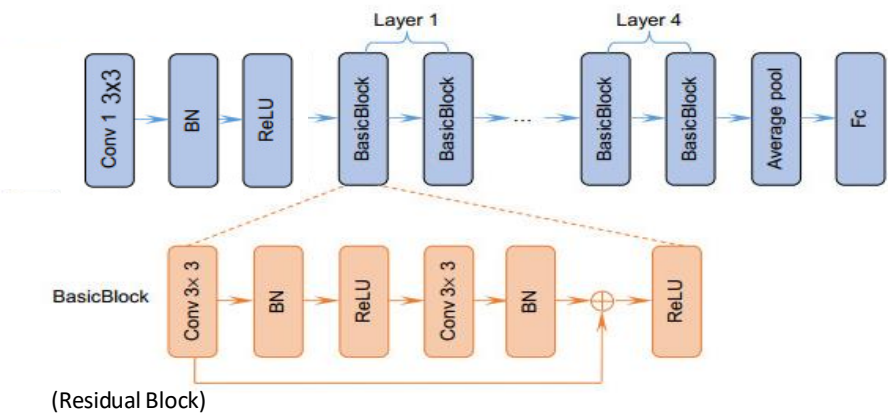


2.2 (10%) Show the Architecture of ResNet18 Model

➢ The output structure of ResNet18 model (出題：Ian)

```
ResNet(
  (conv1): Conv2d(3, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
  (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (relu): ReLU(inplace=True)
  (layer1): Sequential(
    (0): BasicBlock(
      (conv1): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (relu): ReLU(inplace=True)
      (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    )
    (1): BasicBlock(
      (conv1): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (relu): ReLU(inplace=True)
      (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    )
  )
        ⋮
  (avgpool): AdaptiveAvgPool2d(output_size=(1, 1))
  (fc): Linear(in_features=512, out_features=10, bias=True)
)
```

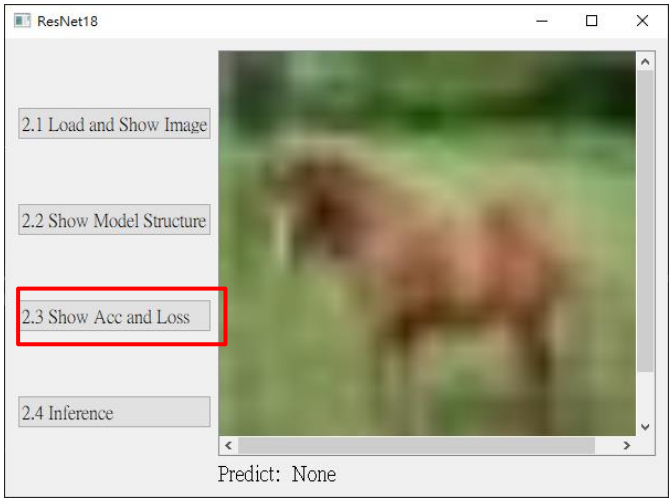
Basic Block 含有這些，並不是順序。順序需要自己實作(可參考下面的圖)。

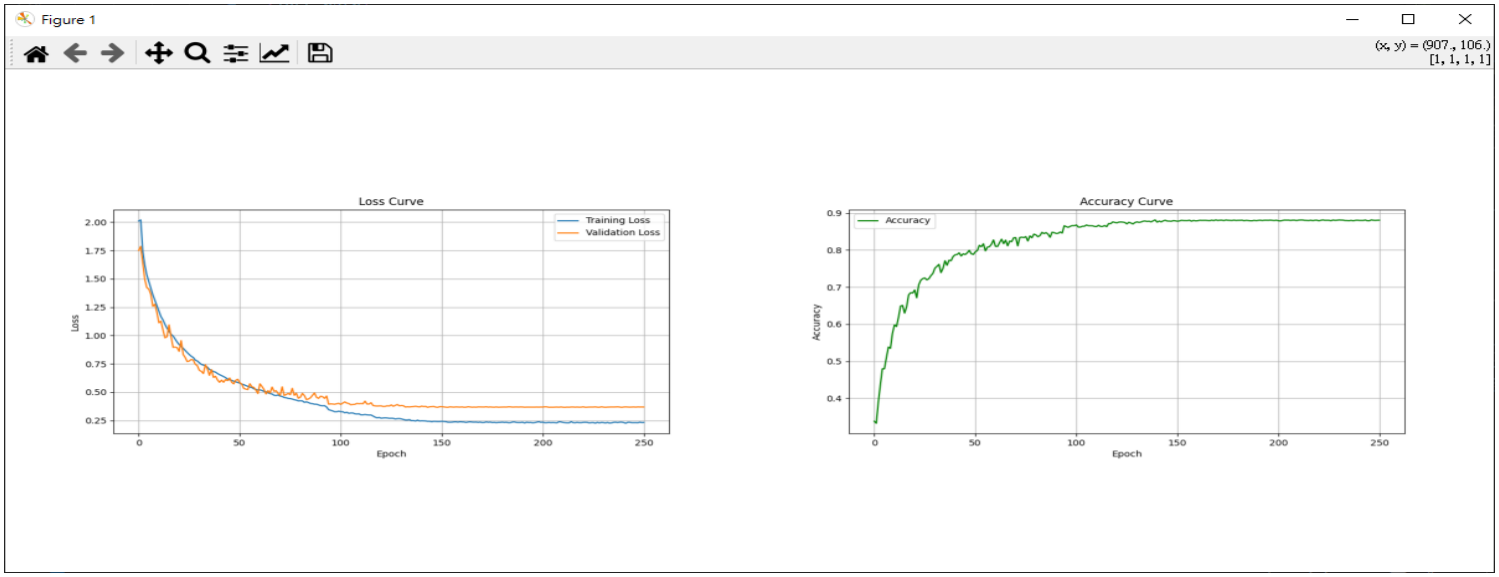


2.3 (15%) Show Training/Validating Loss Curve and Accuracy Curve

(出題：Ian)

1. At home:
   1) Download the datasets from torchvision.dataset, and split training and validation dataset.
   2) Training and validating ResNet-18 at home and record the training/validation loss and accuracy curve for each epoch.
   3) Use matplotlib.pyplot.plot() to create a line chart for the training and validation loss and accuracy values then save the figure.
   4) Save weight file with highest validation accuracy.
2. When the demo:
   1) Click the button "2.3 Show Acc and Loss"
   2) Show the saved figure of Training/Validating loss and accuracy.





The output of loss curve and accuracy curve

2.4 (15%) Use the Trained Model with Highest Validation Accuracy to Run Inference, Show the Probability Distribution and Class Label

(出題：Ian)

1. When the demo:
   1) Load the trained model with highest validation accuracy.
   2) Resize the image to 32x32x3c (RGB)
   3) Click the button "2.4 Inference" to run inference on the image you selected.
   4) Note that there is a special image in the inference dataset that is not included in the 10 classes of CIFAR-10. You should set a threshold (adjust it yourself) to determine whether the image belongs to 10 classes in the CIFAR-10 dataset.
      - Show the predicted class label and confidence score on GUI.
      - Show the probability distribution of model predictions using a histogram.

      ➔ Hint: If the max probability < threshold, the image is not in 10 classes.

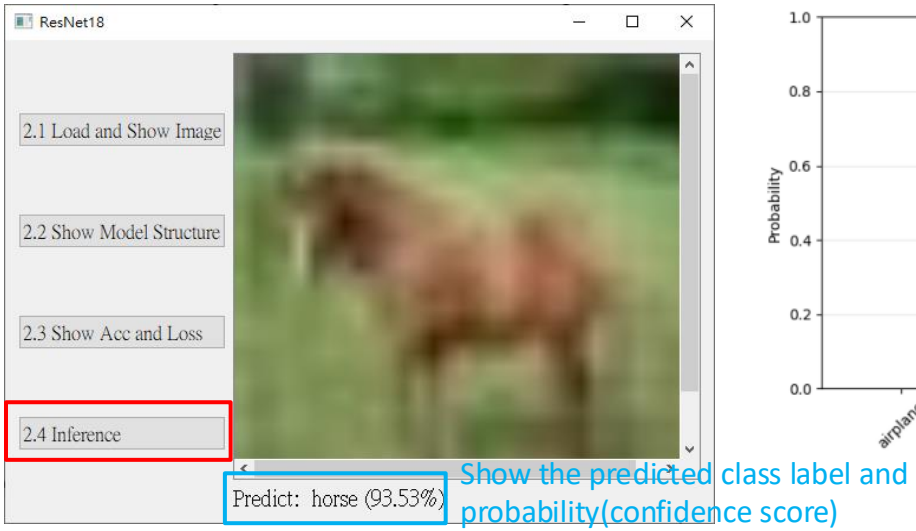
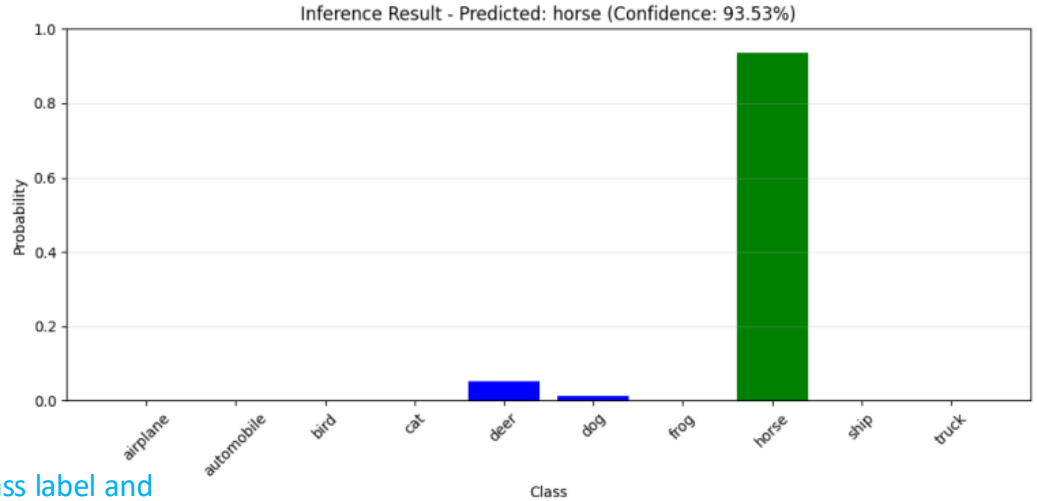


Image in 10 classes of CIFAR-10



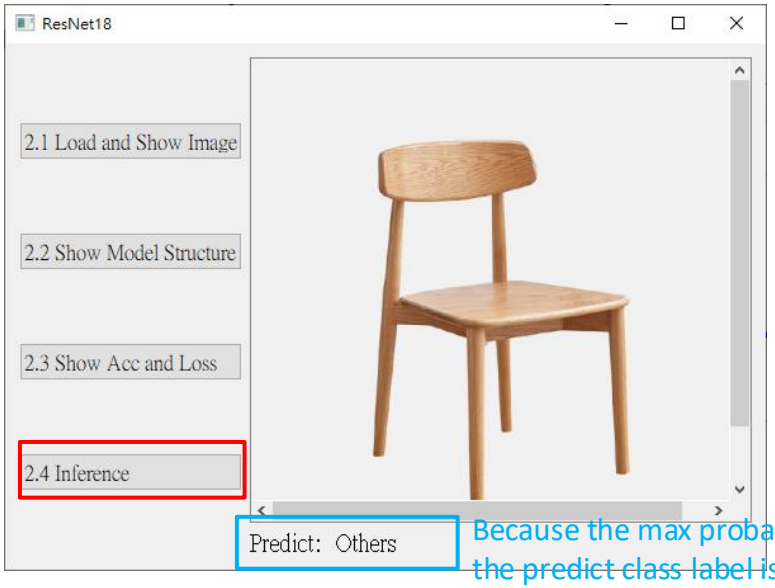
The output of probability distribution

2.4 (15%) Use the Trained Model with Highest Validation Accuracy to Run Inference, Show the Probability Distribution and Class Label

(出題：Ian)

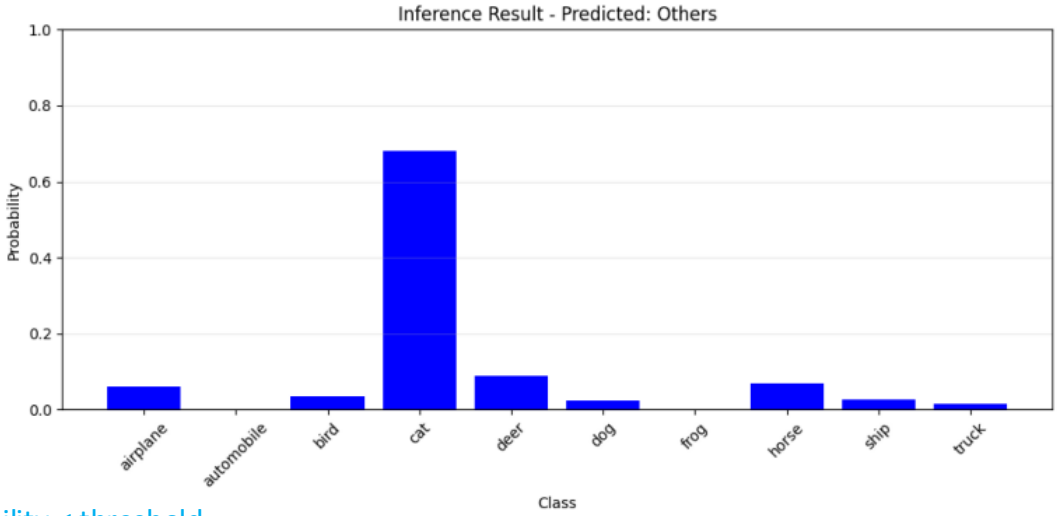
1. When the demo:
   1) Load the trained model with highest validation accuracy.
   2) Resize the image to 32x32x3c (RGB)
   3) Click the button "2.4 Inference" to run inference on the image you selected.
   4) Note that there is a special image in the inference dataset that is not included in the 10 classes of CIFAR-10. You should set a threshold (adjust it yourself) to determine whether the image belongs to 10 classes in the CIFAR-10 dataset.
      - Show the predicted class label and confidence score on GUI.
      - Show the probability distribution of model predictions using a histogram.

      ➔ Hint: If the max probability < threshold, the image is not in 10 classes.



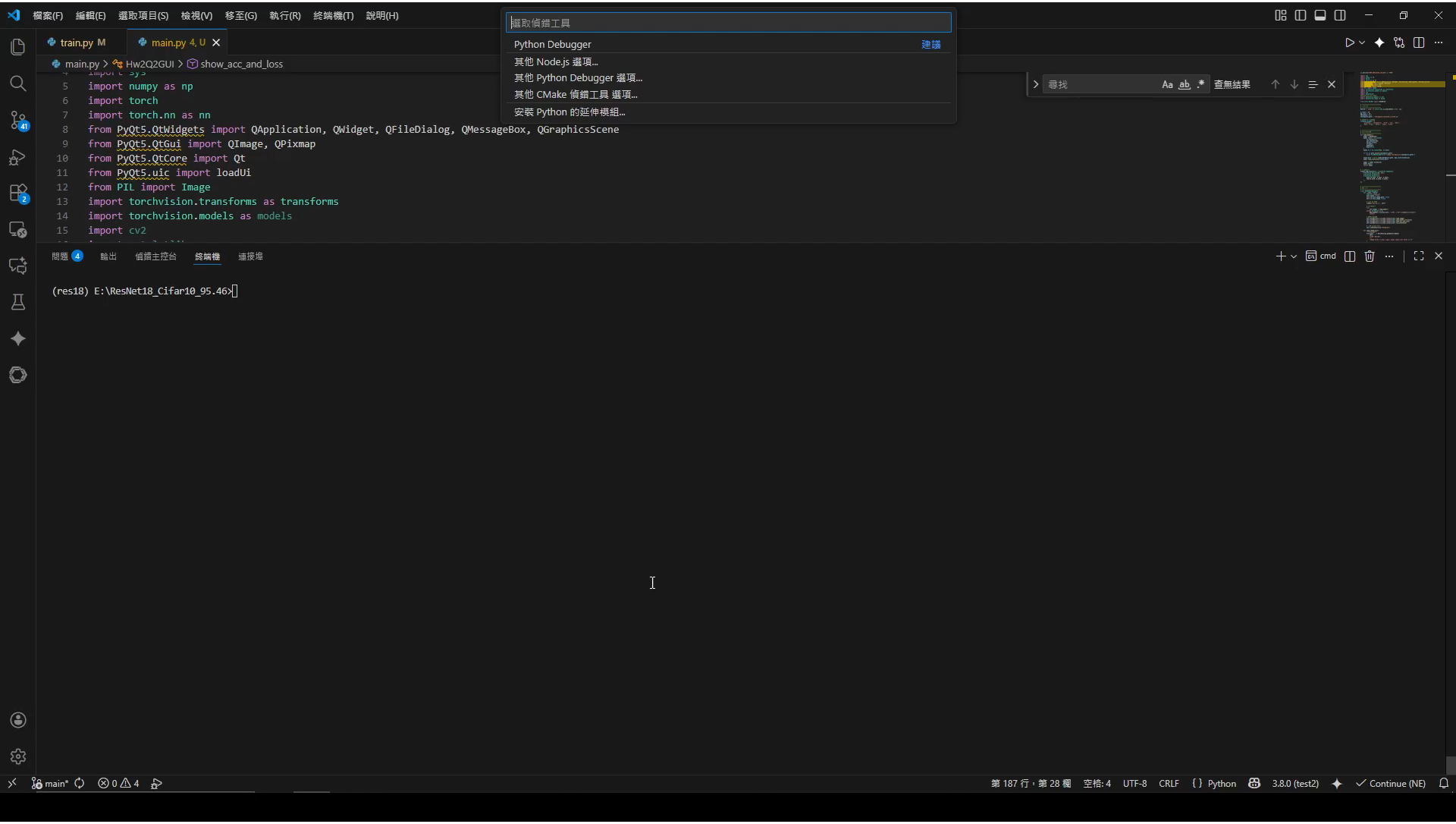
Because the max probability < threshold, the predict class label is "Others"

Image not in 10 classes of CIFAR-10



The output of probability distribution

2. Train a CIFAR-10 Classifier Using ResNet18

(出題：Ian)

- This is an example for demonstration.